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# AIC8800 BT\_TEST

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RF\_TEST

V4.0

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AIC Semiconductor (Shanghai) CO., Ltd.



| company | AIC Semiconductor (Shanghai) CO., Ltd. |                      |
|---------|----------------------------------------|----------------------|
|         | Data                                   | Release note         |
| V1.0    | 2022/4/13                              |                      |
| V2.0    | 2023/12/22                             | Add LE RX command    |
| V3.0    | 2024/4/19                              | Update command       |
| V4.0    | 2024/7/30                              | Add TX power command |

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## What kind of Linux can be set up on a PC?

3.10 < linux kernel version < 6.14 (AIC8800D80X2 supports Linux kernel versions within the range of 3.10 to 6.14)

for example:

Ubuntu version: 18.04, linux kernel version: 5.4

```
aic@aic:~$ uname -a
Linux aic 5.4.0-150-generic #167~18.04.1-Ubuntu SMP Wed May 24 00:51:42 UTC 2023 x86_64 x86_64 x86_64
GNU/Linux
aic@aic:~$
```

Ubuntu version: 20.04, linux kernel version: 5.15

Ubuntu version: 22.04, linux kernel version: 6.1

Ubuntu version: 24.04, linux kernel version: 6.11

## How to set up AIC Linux Driver for PC

### #PC Ubuntu Linux support make and test

#### Step one, copy the firmware

Enter the directory where the firmware is stored android\_driver/USB/driver\_fw/fw/

Copy the folder aic8800D80N to the directory /lib/firmware

sudo cp aic8800D80N /lib/firmware -r

```
aic@aic:~$
aic@aic:~$ cd android_driver/USB/driver_fw/fw/
aic@aic:~/android_driver/USB/driver_fw/fw$ sudo cp aic8800D80N/ /lib/firmware/ -r
aic@aic:~/android_driver/USB/driver_fw/fw$
```

#### Step two Uninstall the pre-installed bt\_test on the computer before loading bt\_test.

```
aic@aic:~$
aic@aic:~$ uname -a
Linux aic 5.15.0-139-generic #149~20.04.1-Ubuntu SMP Wed Apr 16 08:29:56 UTC 2025 x86_64 x86_64 x86_64
GNU/Linux
aic@aic:~$ cd /lib/modules/5.15.0-139-generic/kernel/drivers/bluetooth/
aic@aic:/lib/modules/5.15.0-139-generic/kernel/drivers/bluetooth$ sudo rmmod btusb
aic@aic:/lib/modules/5.15.0-139-generic/kernel/drivers/bluetooth$ sudo mv btusb.ko btusb.kobak
aic@aic:/lib/modules/5.15.0-139-generic/kernel/drivers/bluetooth$
aic@aic:/lib/modules/5.15.0-139-generic/kernel/drivers/bluetooth$
```

#### Step three compile and install bt\_test tools

```
aic@aic:~$
aic@aic:~$ cd android_driver/tools/aicrf_test/
aic@aic:~/android_driver/tools/aicrf_test$ make
gcc -c wifi_test.c -o wifi_test.o
gcc wifi_test.o -o wifi_test
gcc -c bt_test.c -o bt_test.o
gcc bt_test.o -lpthread -o bt_test
aic@aic:~/android_driver/tools/aicrf_test$ sudo make install
sudo cp wifi_test /sbin
sudo cp bt_test /sbin
aic@aic:~/android_driver/tools/aicrf_test$
```

#### Step four, compile and load the driver

Enter the android\_driver/USB/driver\_fw/driver/aic8800 directory and execute the make command

Execute commands in the aic8800 directory

sudo insmod aic\_load\_fw/aic\_load\_fw.ko

```
aic@aic:~$
aic@aic:~$ cd ~/android_driver/USB/driver_fw/drivers/aic8800/
aic@aic:~/android_driver/USB/driver_fw/drivers/aic8800$ make
make -C /lib/modules/5.15.0-139-generic/build M=/home/aic/android_driver/USB/driver_fw/drivers/aic8800
ARCH=x86_64 CROSS_COMPILE= modules
make[1]: warning: jobserver unavailable: using -j1. Add '+' to parent make rule.
make[1]: Entering directory '/usr/src/linux-headers-5.15.0-139-generic'
make[3]: warning: -j4 forced in makefile: resetting jobserver mode.
make[1]: Leaving directory '/usr/src/linux-headers-5.15.0-139-generic'
aic@aic:~/android_driver/USB/driver_fw/drivers/aic8800$ sudo insmod aic_load_fw/aic_load_fw.ko
aic@aic:~/android_driver/USB/driver_fw/drivers/aic8800$
```

#### Step five, compile and load the driver

Enter the android\_driver/USB/driver\_fw/driver/aic\_btusb directory and execute the make command

Execute commands in the aic\_btusb directory

sudo insmod aic\_btusb.ko



```
al@al:~$  
al@al:~$ cd android_driver/USB/driver_fw/drivers/aic_btusb/  
al@al:~/android_driver/USB/driver_fw/drivers/aic_btusb$ make  
make -C /lib/modules/5.15.0-139-generic/build M=/home/aic/android_driver/USB/driver_fw/drivers/aic_btusb ARCH=x86_64 CROSS_COMPILE= modules  
make[1]: Entering directory '/usr/src/linux-headers-5.15.0-139-generic'  
make[1]: Leaving directory '/usr/src/linux-headers-5.15.0-139-generic'  
al@al:~/android_driver/USB/driver_fw/drivers/aic_btusb$ sudo insmod aic_btusb.ko  
[sudo] password for aic:  
al@al:~/android_driver/USB/driver_fw/drivers/aic_btusb$
```

Then insert the usb board

#### Step six

```
sudo chmod 777 /dev/aicbt_dev
```

```
bt_test -s usb
```

#### Step seven start testing

Start inputting the bt\_test command test in another terminal.

```
al@al:~$ cd ~/android_driver/USB/driver_fw/drivers/aic_btusb/  
al@al:~/android_driver/USB/driver_fw/drivers/aic_btusb$ sudo insmod aic_btusb.ko  
[sudo] password for aic:  
Sorry, try again.  
[sudo] password for aic:  
al@al:~/android_driver/USB/driver_fw/drivers/aic_btusb$ cd -  
al@al:~$  
al@al:~$ sudo chmod 777 /dev/aicbt_dev  
al@al:~$ sudo chmod 777 /dev/aicbt_dev  
al@al:~$ bt_test -s usb  
bt_hci_tool server setting  
use USB  
bt dev open successful  
to test hci reset for bt device  
COMMAND(4):  
01 03 0C 00  
EVENT(7):  
04 0E 04 05 03 0C 00  
bt device test ok  
hci rcv thread ready (nil)
```

```
Usage:  
-s <interface> <params>  
Select the tool service:  
ex. "bt_test -s uart 1500000 /dev/ttyS0" or "bt_test -s usb" or "bt_test -s wlan wlan0"  
-c <hci_cmd> Send an HCI command to the selected interface  
-v Display Release Date  
  
Commands:  
bt_start [<trx> <chan> <pkt> ...] = Start Bluetooth tx test  
  <trx> (man) = tx | rx  
  <chan> (man) = 0 ~ 78 (hopen=off) | 255 (hopen=on)  
  <pkt> (man) = dh1 | dh3 | dh5 | 2dh1 | 2dh3 | 2dh5 | 3dh1 | 3dh3 | 3dh5  
  <pattern> (opt) = pn9 (default) | pn15 | h00 | hff | hf0 | hf8 | haa | h55  
  <len> (opt) = dec value (default: pkt's max length)  
  <whiten> (opt) = on | off (default)  
  <addr> (opt) = 0a 1c 6b c6 96 7e (default)  
bt_stop [<trx>] = Stop Bluetooth tx test  
  <trx> (man) = tx | rx  
ble_start [<trx> <chan> <pkt> ...] = Start BLE tx test  
  <trx> (man) = tx | rx  
  <chan> (man) = 0 ~ 39 (hopen=off) | 255 (hopen=on)  
  <pkt> (man) = 1m | 2m | s2 | s8  
  <pattern> (opt) = pn9 (default) | pn15 | h00 | hff | hf0 | hf8 | haa | h55  
  <len> (opt) = dec value (default: 255)  
ble_stop [<trx>] = Stop BLE tx test  
  <trx> (man) = tx | rx  
tx_tone [<chan> ...] = Start a Single-Carrier Bluetooth test  
  <chan> (man) = 0 ~ 78  
  <pw> (opt) = 0 ~ 7 (default: 0x07)  
tx_tonestop = Stop Bluetooth Single-Carrier Test  
get_txpw = Get tx power  
set_txpw [<pw>] = Set tx power  
  <pw> (man) = 0 ~ 0x0f (default: 0x0f)  
bt_reset = reset BT  
bt_dut <i/o> = bt_dut /bt_dut 1 (dut en & scan on) | bt_dut 0 (scan off)  
set_pwrfst [<pwrfst>] = Set bt tx power offset  
  <pwrfst> (man) = -7 ~ 7
```

BT test case:

BT\_TX:

BR DH5 PRBS9

```
bt_test bt_start tx 0 dh5 \ 2402MHz  
bt_test bt_start tx 39 dh5 \ 2441MHz  
bt_test bt_start tx 78 dh5 \ 2480MHz
```

BR 2DH5 PRBS9

```
bt_test bt_start tx 0 2dh5 \ 2402MHz  
bt_test bt_start tx 39 2dh5 \ 2441MHz  
bt_test bt_start tx 78 2dh5 \ 2480MHz
```

BR 3DH5 PRBS9

```
bt_test bt_start tx 0 3dh5 \ 2402MHz  
bt_test bt_start tx 39 3dh5 \ 2441MHz  
bt_test bt_start tx 78 3dh5 \ 2480MHz
```

BT\_RX:

BR DH5 PRBS9

```
bt_test bt_start rx 0 dh5 \ 2402MHz  
bt_test bt_start rx 39 dh5 \ 2441MHz  
bt_test bt_start rx 78 dh5 \ 2480MHz
```



## BR 2DH5 PRBS9

bt\_test bt\_start rx 0 2dh5 \\ 2402MHz  
bt\_test bt\_start rx 39 2dh5 \\ 2441MHz  
bt\_test bt\_start rx 78 2dh5 \\ 2480MHz

## BR 3DH5 PRBS9

bt\_test bt\_start rx 0 3dh5 \\ 2402MHz  
bt\_test bt\_start rx 39 3dh5 \\ 2441MHz  
bt\_test bt\_start rx 78 3dh5 \\ 2480MHz

## BLE test case:

## BLE\_TX:

bt\_test ble\_start tx 0 1m \\ TX BLE 1M 2402MHz PRBS9  
bt\_test ble\_start tx 0 2m \\ TX BLE 2M 2402MHz PRBS9  
bt\_test ble\_start tx 0 s2 \\ TX BLE S2 2402MHz PRBS9  
bt\_test ble\_start tx 0 s8 \\ TX BLE S8 2402MHz PRBS9

## BLE\_RX

bt\_test ble\_start rx 0 1m \\ RX BLE 1M 2402MHz PRBS9  
bt\_test ble\_start rx 0 2m \\ RX BLE 2M 2402MHz PRBS9  
bt\_test ble\_start rx 0 s2 \\ RX BLE S2 2402MHz PRBS9  
bt\_test ble\_start rx 0 s8 \\ RX BLE S8 2402MHz PRBS9

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## bt\_test instructions for use

### 1、Preparation

```
svc wifi disable  
chmod -R 777 /dev/aicbt_dev (USB) /dev/tty*(SDIO)  
rmmod aic8800_fdrv  
rmmod aic_load_fw  
insmod /vendor/lib/modules/aic_load_fw.ko
```

### 2、bt\_test help

<-s> to be tool service. ex. "bt\_test -s uart 115200 /dev/ttyS0" or "bt\_test -s usb" or "bt\_test -s wlan wlan0"  
<-c> to send hci cmd to interface.  
<-w> to send wlan cmd to interface.

### 3、bt\_test -s uart 1500000 /dev/ttyS0 &

Take uart bt as an example, open bt dev, the baud rate is set to the bsp driver, and /dev/ttyS0 is the uart bt interface.

Open the test, and after entering the following sentence normally, it will open and display successful and return to EVENT printing.

## 4. BT fixed frequency command

### 4.1 BT TX test

(1) bt\_test bt\_start par0 par1 par2 par3 par4 par5 par6

Note: par3 par4 par5 par6 can not be written by default

Eg: TX 2402MHz DH1:

default:

bt\_test bt\_start tx 0 dh1 \ BR 2402MHz DH1 length 27 PRBS9 whiten off addr 0a 1c 6b c6 96 7e

optional:

eg: bt\_test bt\_start tx 0 dh1 haa 27 on xx xx xx xx xx xx

\ BR 2402MHz DH1 10101010 length 27 whiten on addr xx xx xx xx xx xx

| param | optional | Name    | Value range                               |
|-------|----------|---------|-------------------------------------------|
| Par0  | man      | trx     | tx<br>rx                                  |
| Par1  | man      | chan    | 0~78(hopen=off)<br>255(hopen=on)          |
| Par2  | man      | pkt     | dh1 dh3 dh5                               |
|       |          |         | 2dh1 2dh3 2dh5                            |
|       |          |         | 3dh1 3dh3 3dh5                            |
| Par3  | opt      | pattern | pn9((PRBS9)(default))                     |
|       |          |         | pn9 PRBS9 hff 11111111                    |
|       |          |         | pn15 PRBS15 h00 00000000                  |
|       |          |         | haa 10101010 hf0 11110000                 |
|       |          |         | h55 01010101 h0f 00001111                 |
| Par4  | opt      | length  | decimal value (default: pkt's max length) |
|       |          |         | dh1 27 dh3 183 dh5 339                    |
|       |          |         | 2dh1 54 2dh3 367 2dh5 679                 |
|       |          |         | 3dh1 83 3dh3 552 3dh5 1021                |
| Par5  | opt      | whiten  | off (default)<br>on                       |
| Par6  | Opt      | addr    | default: 0a 1c 6b c6 96 7e                |

(2) bt\_test bt\_stop par0

| param | optional | Name | Value range |
|-------|----------|------|-------------|
| par0  | man      | trx  | tx<br>rx    |

BT\_RX stop after getting RX result:

printf("rx = %d, per = %d, ber = %.3f%%", sync\_ok, int(pkt\_rx\_ok/sync\_ok), bit\_err/bit\_cnt\*100.0)

firmware return event contain: sync\_ok, pkt\_rx\_ok, bit\_cnt, bit\_err

bit\_cnt = sync\_ok \* len \* 8

## 5、BLE test

### 5.1 TX, LE Transmitter Test command

(1) bt\_test ble\_start par0 par1 par2 par3 par4

| param | optional | Name    | Value range                              |
|-------|----------|---------|------------------------------------------|
| Par0  | man      | trx     | tx<br>rx                                 |
| Par1  | man      | chan    | 0~39(hopen=off)<br>255(hopen=on)         |
| Par2  | man      | pkt     | 1M      2M      S2      S8               |
| Par3  | opt      | pattern | pn9((PRBS9)(default))                    |
|       |          |         | pn9      PRBS9      hff      11111111    |
|       |          |         | pn15      PRBS15      h00      00000000  |
|       |          |         | haa      10101010      hf0      11110000 |
|       |          |         | h55      01010101      h0f      00001111 |
| Par4  | opt      | length  | 1~255 (default: 255)                     |

(2) bt\_test ble\_stop par0 par1 par2 par3 par4

| param | optional | Name | Value range |
|-------|----------|------|-------------|
| par0  | man      | trx  | tx<br>rx    |

BLE\_RX stop received RX result:  
return event contain: pkt\_rx\_ok



## 6、BT single tone

bt\_test tx\_tone par0 par1

| param | optional | Name | Value range                     |
|-------|----------|------|---------------------------------|
| Par0  | man      | chan | 0~78                            |
| Par1  | opt      | pwr  | 00-7f (default :7f (hex value)) |

BT single tone test case:

bt\_test tx\_tone 0 \\2402MHz single tone

bt\_test tx\_tone 39 \\2441MHz single tone

bt\_test tx\_tone 78 \\2480MHz single tone

bt\_test tx\_tonestop \\tone stop

## 7、BT DUT fixed frequency commands

bt\_test bt\_dut 1 \\start scan

bt\_test bt\_dut 0 \\Turn off scan when the meter is connected to the device

## 8、BT TX POWER read and write

bt\_test get\_txpwr \\ Reading TX power gain level

bt\_test set\_txpwr par0 \\ Setting TX power gain level

| param | optional | Name             | Value range |
|-------|----------|------------------|-------------|
| Par0  | man      | Power gain level | 0~0x7f      |